

A three-step chip-firing counterexample to Conjecture 00000001237

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October 9, 2026

Abstract

On the directed triangle with three unit-length arcs, a single chip travels around the cycle. Its least return period is three, whereas the least common multiple of the arc lengths is one. The counterexample has exactly one legal firing at each step, so it applies to both parallel chip firing and one-vertex-at-a-time chip firing.

1 The proposed formula

The source defines the chip-firing period to be the length of the recurrent oscillation and asserts that, on a directed cycle, this period equals the least common multiple of the arc lengths. We take the ordinary directed cycle on the labeled vertices $V = \mathbb{Z}/3\mathbb{Z}$, with arcs

$$0 \longrightarrow 1, \quad 1 \longrightarrow 2, \quad 2 \longrightarrow 0.$$

Every arc has length one. This is also a valid instance of the formula if lengths are treated as explicit positive integral edge data. The graph is strongly connected and each vertex has indegree and outdegree one.

2 The chip-firing dynamics

A configuration is a function $c : V \rightarrow \mathbb{N}$. A vertex v can fire when $c(v)$ is at least its outdegree. Firing subtracts that outdegree from $c(v)$ and sends one chip along each outgoing arc. In our graph a firing moves one chip from v to $v + 1$. Let e_v denote the configuration with one chip at v and zero elsewhere.

Theorem 1. *Starting from e_0 , chip firing has least positive return period three. The least common multiple of the arc lengths is one. Hence the proposed period formula is false.*

Proof. In e_v , vertex v is legal and both other vertices are illegal. Its unique legal firing produces e_{v+1} . The entire orbit is therefore

$$(1, 0, 0) \longmapsto (0, 1, 0) \longmapsto (0, 0, 1) \longmapsto (1, 0, 0).$$

The first three configurations are distinct. The first positive return is thus at time three. Exactly one vertex is legal at every state in this orbit, so simultaneous firing of all legal vertices yields the same transitions as sequential firing. On the other hand, the three arc lengths are 1, 1, 1, whose least common multiple is 1. Since $3 \neq 1$, the asserted equality fails. $\square$

Counting convention. The period here counts firings (equivalently, parallel time steps) until the same configuration returns on the same fixed vertex set. The configurations are not identified under graph rotations. The source asks for the length of a recurrent oscillation, not for the number of completed laps or the number of firings of one designated vertex. Under the stated configuration-return notion the cycle above has period three.

3 Formal verification

`lean/Main.lean` defines the actual finite arc set, its lengths, outdegrees, configurations, legal vertices, and chip-firing updates. The parallel update subtracts the chips sent by a legal vertex and sums the chips received along incoming arcs; it is not defined by declaring a three-cycle. A separate `fire` function implements the usual single-vertex update.

- `arc_count`, `one_outgoing_arc`, and `one_incoming_arc` check the actual directed triangle. `directed_cycle_strongly_connected` gives paths of length at most two between all vertex pairs.
- `unique_legal_vertex` and `one_chip_transition` prove legality and the exact update. `parallel_equals_unique_legal_firing` relates the parallel and sequential conventions in this orbit.
- `actual_period` computes Mathlib’s `Function.minimalPeriod` as three. `arc_lengths_lcm` computes the finite-set lcm as one. `conjectured_period_formula_false` is their inequality.

Finite computations are checked by Lean’s kernel through ordinary `decide`. No auxiliary numerical script is required. No unproved placeholders, custom axioms, or native evaluation are used as proofs.

Reproduction. In `lean/`, run:

```
lake build
lake env lean -DwarningAsError=true Main.lean
```

Lean 4.19.0 and the exact Mathlib revision are pinned. Run `tectonic main.tex` from the submission directory for the PDF. The exact bilingual conjecture is in `SOURCE.md`.

References

- [1] The Last Math Competition, Conjecture 00000001237.
- [2] The Mathlib Community, Lean 4 mathematical library, revision c44e0c8ee63c.